Human Motion Animation Generation

Autoregressive Context Encoding & Spatial-Temporal Flow Matching

Your Name

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Problem Statement

Goal: Generate highly realistic 3D human motion animations conditioned on text descriptions.

Context & Dataset:

- Utilizing the HumanML3D dataset format.
- Translating text prompts (e.g., "A person is walking forward") into a sequence of 22-joint articulated poses.

Previous Iterations

Early Architectures (To be expanded)

- *Details of past approach 1 (e.g., GRU-based History Encoder)*
- *Details of past approach 2 (e.g., Latent-only state representations)*

Animation Demonstration

[Placeholder for previous iteration MP4]

System Overview

A text-conditioned autoregressive motion generator dividing generation into 4 stages per frame:

1. **Encode**: Summarize 271D motion history with text conditioning.
2. **Predict**: Denoise next-frame state into a reduced 68D representation via Flow Matching.
3. **Convert**: Map 68D state back into absolute global joint positions.
4. **Accumulate**: Derive new 271D features and append to explicit position history.

Data Representations

Bridging motion geometry and model efficiency:

271D Full Frame (Encoder & Dataset)

- Root state (3), Root-Invariant Coords (66), 6D Rotations (132), Local Vel (66), Contacts (4).

68D Reduced Target Space (Flow Matching)

- Model-friendly target for flow solver.
- Root height/velocity/yaw (5) + 21 Non-root joints RIC (63).

1. Motion History Encoder

Causal Temporal Attention Network

- **Input:** Variable-length history window $X \in \mathbb{R}^{B \times T \times 271}$.
- **Text Conditioning:** Pooled CLIP text embeddings modulate every timestep via shift/scale.
- **Mechanism:** Causal temporal self-attention with RoPE + gated MLP blocks.
- **Output:** 22 joint tokens supplying context without forcing the predictor to model full spatiotemporal attention.

2. Flow Matching Predictor

Spatial Transformer Engine

- **Inputs (per token):** Noisy 68D features + History Context + 257D causal previous-frame features.
- **Conditioning:** AdaLN (Time t embedding + CLIP Text projection).
- **Structure Prior:** Kinematic chain embeddings define skeletal hierarchy (depth + chain ID).
- **Action:** Predicts velocity fields to regress the denoised 68D state.

Training & Inference

Training Loop

- Incremental flow loss optimization.
- Parallel sequences using $X_{:-1}$ to predict $X_{1:}$.
- Uses interpolated state $x_t = t \cdot x_1 + (1 - t) \cdot x_0$.

Inference (Position-First)

- Integrated using end-biased Heun ODE solver.
- Absolute positions act as the primary rolling state.
- Option to utilize Forward Kinematics (FK) consistency for skeletal stability.

Generation Comparisons

Ground Truth
Placeholder: gt_demo.mp4

Full Autoregressive Rollout
Placeholder: rollout_demo.mp4

Teacher-Forced Comparisons

1-Step Teacher Force
Placeholder: tf_1step.mp4

Masked Teacher Force
Placeholder: tf_masked.mp4

Conclusion

- Successfully separated causal temporal accumulation and spatial-temporal next-frame denoising.
- Retained geometric fidelity by using a position-first recursive inference state.
- Future Work: Enable scheduled sequence rollouts during training and cache state optimizations.

Questions?